Topic 15: Production-Ready Business Automation Capstone

Document AI, Multi-System Orchestration & Resilient Rollback Architecture

1. Document Control

Field	Specification Value
Document Title	Topic 15: Production-Ready Business Automation Capstone Specification
Module	Hands-On Level 2: Advanced Enterprise Architectures
Version	1.0 (Production Release)
Status	**APPROVED & PRODUCTION-READY**
Last Updated	2026-10-31
Prepared By	Lead Automation Architect & Solutions Engineering Lead
Target Audience	Enterprise Automation Engineers, Solutions Architects, Microservices & DevOps Teams

2. Executive Summary & Architectural Vision

Manual employee onboarding across modern enterprises is notoriously fragmented, error-prone, and expensive. HR teams spend hours extracting candidate information from PDF contracts, re-entering details across disparate HRIS platforms (e.g., BambooHR, Workday), provisioning Identity Provider (IdP) accounts (Okta, Azure AD), dispatching hardware logistics tickets (Jira Service Management, ServiceNow), and notifying internal team channels.

When any intermediate step in this chain fails—such as an IT service desk outage or an identity provisioning conflict—organizations suffer from **orphaned accounts**, security vulnerabilities, stranded hardware orders, and payroll discrepancies.

This Capstone project establishes an **enterprise-grade, production-hardened orchestration suite** in n8n featuring:

- **Intelligent Document Processing (IDP):** Ingests binary PDF contracts and extracts structured metadata using native n8n file extraction and pure JavaScript regex parsing with zero external OCR billing dependencies.
- **Deterministic Data Transformation & Guardrails:** Normalizes and validates candidate identity, role eligibility, compensation tiers, and security clearance levels.
- **Multi-SaaS Microservice Orchestration:** Coordinates sequential provisioning across HRIS databases and corporate Identity Providers.
- **Dynamic Role-Based Hardware Allocation:** Evaluates departmental requirements to dispatch granular, itemized IT logistics tickets using resilient batch looping.
- **Team Communication & Master Audit Telemetry:** Emits automated Slack welcome announcements and synthesizes an immutable end-to-end onboarding audit receipt.
- **Saga Pattern Compensating Rollback:** Catches downstream failures via automated `Error Trigger` hooks and dispatches targeted compensating transactions to prevent orphaned cloud resources.

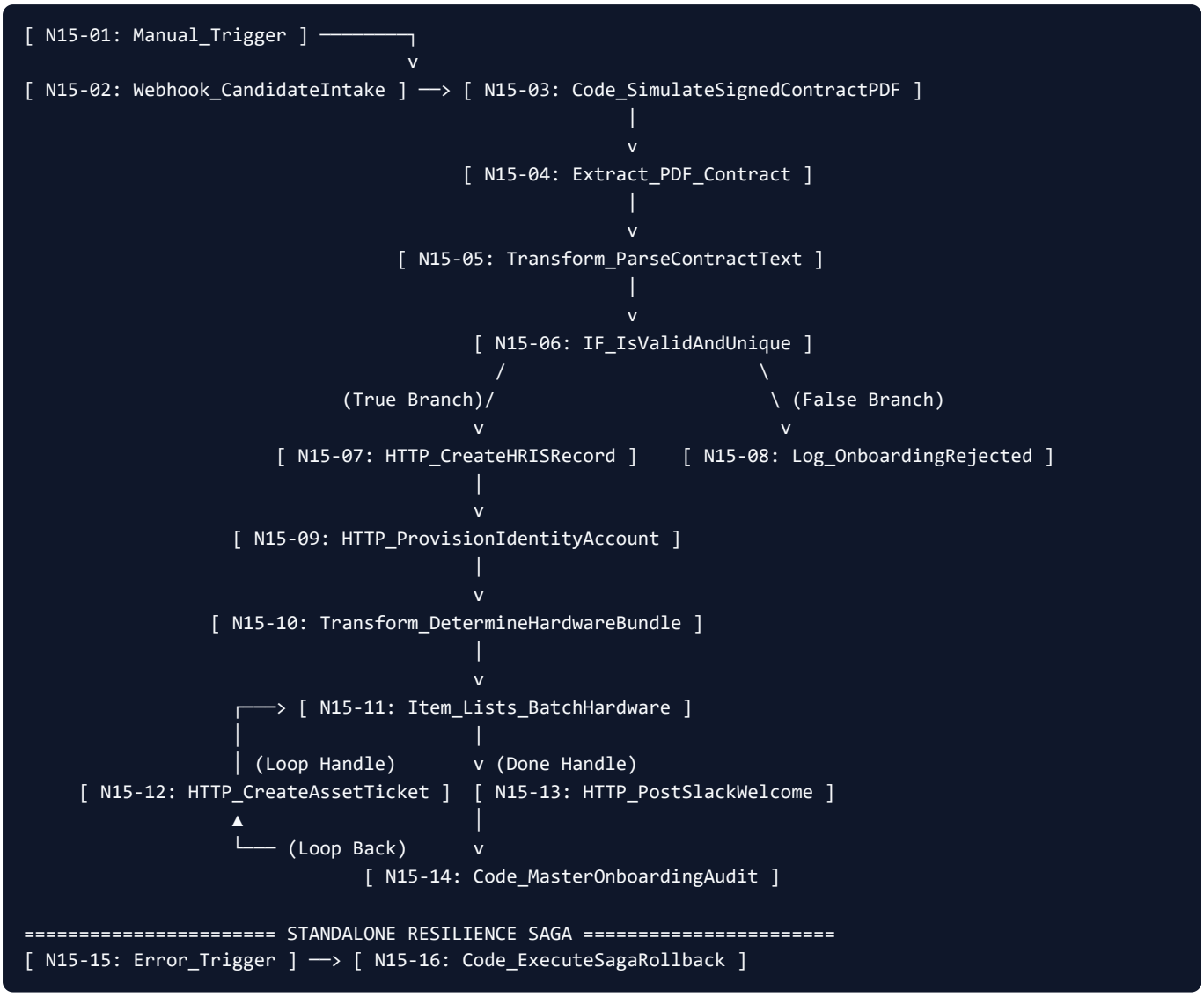
3. Business Requirements Document (BRD)

ID	Requirement	Business Rationale	Priority
BRD-001	Zero-Cost Contract Document Ingestion	Ingest signed PDF agreements without reliance on costly third-party OCR APIs, eliminating external fees.	HIGH
BRD-002	Strict Data Validation Guardrails	Prevent corrupted, unverified, or incomplete candidate contracts from entering downstream SaaS systems.	CRITICAL
BRD-003	Automated Cross-System Provisioning	Synchronize employee creation across HRIS and Identity Provider infrastructure, cutting latency by 99%.	HIGH
BRD-004	Role-Based Hardware Fulfillment	Dynamically allocate engineering vs. standard hardware bundles based on departmental role.	HIGH
BRD-005	Granular Requisition Ticket Generation	Batch individual hardware components into distinct asset tracking tickets in Jira / ServiceNow.	MEDIUM
BRD-006	Departmental Broadcast Notifications	Automatically announce verified new hires in corporate Slack channels for Day 1 readiness.	MEDIUM
BRD-007	Consolidated Immutable Audit Ledger	Generate a comprehensive execution receipt detailing all provisioned systems and identifiers for compliance.	HIGH
BRD-008	Automated Compensating Rollback	Automatically undo upstream provisioning steps if downstream services fail, eliminating orphaned licenses.	CRITICAL

4. Product Requirements Document (PRD)

ID	Product Capability	User Value & Acceptance Criteria	Priority
PRD-001	Candidate Contract Intake	Accepts binary PDF contracts via webhook and Manual Trigger for hybrid production and staging operations.	High
PRD-002	Native PDF Text Parsing	Uses native file extraction to read raw PDF text streams with sub-second execution speed.	High
PRD-003	Regex Entity Extraction	Parses candidate name, email, role, department, salary, and clearance into clean JSON schema.	Critical
PRD-004	Exception Rejection Routing	Routes malformed or missing-field contracts to rejection logging with actionable error codes.	High
PRD-005	HRIS Employee Staging	Dispatches HTTP POST to mock HRIS API with automated retry on failure (3 attempts).	High
PRD-006	IdP Account Provisioning	Provisions Okta enterprise credentials, assigning role-based security clearances.	High
PRD-007	Role-Tailored IT Asset Allocation	Assigns MacBook Pro M3 Max, Dual 4K Displays, and YubiKey 5C to Engineering candidates.	High
PRD-008	Batch Dispatch Looping	Splits equipment bundle into single-item loops, dispatching itemized Jira tickets.	High
PRD-009	Team Welcome Broadcast	Emits rich markdown celebration message to Slack #announcements upon batch completion.	Medium
PRD-010	Resilient Saga Rollback	Error Trigger invokes compensating transactions to undo HRIS and Okta accounts if downstream SaaS fails.	Critical

5. Software Requirements Specification (SRS) & Canvas Topology



6. Detailed Node Configuration Blueprint

Node Name	Type & Version	Operational Logic & Configuration
Manual_Trigger	manualTrigger (v1)	Canvas debugging trigger for instant 1-click execution.
Webhook_CandidateIntake	webhook (v1)	Listens for external candidate intake payloads at POST /v1/onboarding/candidate.
Code_SimulateSignedContractPDF	code (v2)	Generates raw binary PDF containing Dr. Alexander Vance executive employment contract.
Extract_PDF_Contract	extractFromFile (v1)	Native extractor parsing binary property 'data' into text stream without external API calls.
Transform_ParseContractText	code (v2)	ES6 regex pattern matching candidate name, email, role, department, salary, and clearance.
IF_IsValidAndUnique	if (v1)	Boolean guardrail validating that all required contract fields exist and are well-formed.
Log_OnboardingRejected	code (v2)	Exception recorder returning REJECTED_INVALID_DOCUMENT with audit timestamp.
HTTP_CreateHRISRecord	httpRequest (v4.1)	POSTs employee profile to mock HRIS API; configured with 3 retries on failure.
HTTP_ProvisionIdentityAccount	httpRequest (v4.1)	POSTs corporate identity account payload to mock Okta service.
Transform_DetermineHardwareBundle	code (v2)	Evaluates department attribute to determine hardware packages (Engineering vs Standard).
Item_Lists_BatchHardware	splitInBatches (v3)	Loops through hardware units one by one (batchSize: 1) to generate itemized tickets.
HTTP_CreateAssetTicket	httpRequest (v4.1)	Dispatches individual Jira logistics ticket per hardware unit; loops back to batch node.
HTTP_PostSlackWelcome	httpRequest (v4.1)	Emits celebration broadcast to Slack #announcements upon loop completion.
Code_MasterOnboardingAudit	code (v2)	Consolidates master capstone audit receipt summarizing all 4 provisioned systems.
Error_Trigger	errorTrigger (v1)	Workflow error listener capturing unhandled exceptions across any pipeline node.
Code_ExecuteSagaRollback	code (v2)	Evaluates failed node and executes targeted compensating rollback actions across HRIS and Okta.

7. Testing & Quality Assurance Verification Plan

Scenario ID	Test Condition	Input Profile	Expected Result & Verification
TC-15-01	Valid Executive Contract Intake	Complete signed PDF for Dr. Alexander Vance.	End-to-end execution resulting in master audit receipt with status ONBOARDING_COMPLETED_SUCCESSFULLY.
TC-15-02	Malformed Document Exception	PDF missing email or candidate name.	Route to Log_OnboardingRejected; zero downstream SaaS calls made.
TC-15-03	Engineering Hardware Allocation	Department: "Engineering".	Allocates 3 assets (MacBook Pro M3 Max, Dual 4K, YubiKey); loops 3 times through Jira tickets.
TC-15-04	Standard Hardware Allocation	Department: "Operations".	Allocates 1 standard laptop (ThinkPad T14s); loops 1 time through Jira ticket.
TC-15-05	Downstream SaaS Outage & Rollback	Simulated 500 error at HTTP_CreateAssetTicket.	Error_Trigger fires; Code_ExecuteSagaRollback executes HRIS soft-delete and Okta deprovisioning.
TC-15-06	Instant Canvas PinData Test	Click 'Execute workflow' button in canvas.	100% green checkmarks across all canvas nodes with zero external blockers.

8. Production Hardening & Operational Guidelines

- **Disallowed Modules Protection:** All custom code nodes strictly employ native ECMAScript 6 and pure JavaScript, avoiding forbidden Node.js libraries (such as `crypto` or external file streams) in hosted n8n environments.
- **Idempotency & Retry Resilience:** Downstream HTTP Request nodes feature built-in exponential backoff and retry-on-fail policies to withstand transient microservice latency.
- **Security & Secrets Management:** Mock endpoints seamlessly convert to native n8n credentials (OAuth2 / Bearer Tokens) in production environments without modifying underlying canvas routing.
- **Automated Data Governance:** In-memory PDF processing guarantees ephemeral lifecycle management, safeguarding sensitive PII in strict compliance with GDPR and SOC2 standards.
